

ESN–A2 Stochastic-Volatility Model: Architecture and hyperparameters tuning

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July 14, 2026

Abstract

This technical note is organised as follows:

1. **Model specification.** introduces the complete ESN–A2 stochastic-volatility model, including its architecture, the data-generating processes, and the score function used for calibration.
2. **Calibration framework.** presents the implementation choices required for the calibration procedure.
3. **Scoring methodology.** defines the smooth Gaussian-kernel score that is employed throughout the study.
4. **Model hyperparameters tuning and empirical results.** Section §4 evaluates the proposed architecture through several complementary analyses:
 - (a) a complete grid search over the reservoir dimensions (N_r, N_z) , averaged over three independent random seeds (Section 5.2);
 - (b) the calibrated read-out parameters obtained for each volatility scenario (Section 5.3);
 - (c) the model’s ability to reproduce the target Hurst exponent across four rough- and persistent-volatility scenarios (Section 5.4).
5. **Conclusions.** Finally, it summarises the resulting optimal architecture and outlines the remaining directions for further validation.

Contents

1	The Model	2
1.1	Two parameter classes	2
1.2	ESN architecture	2
1.2.1	The r -layer	2
1.2.2	The z -layer	3
1.2.3	Coupling to the output	3
2	Data Generating Process (DGP) portfolio for the grid search procedure	4
2.1	Score function	5
2.2	Search procedure	5
3	Grid-Matrix Hyperparameter Search: Algorithm	6
4	Implementation Requirements	8
5	Results	9
5.1	The optimal architecture	9
5.2	Grid-matrix sweep	9
5.3	Per-scenario calibrated read-out parameters	10
5.4	Hurst-matching accuracy	11

1 The Model

1.1 Two parameter classes

The pipeline separates two conceptually distinct parameter sets.

- **Hyperparameters** (shared across every DGP scenario, found by an outer CMA-ES search): reservoir sizes N_r, N_z ; z-bank amplitude (**z_strength**); pre-activation coefficients $\beta_{\text{even}}, \beta_{\text{lin}}$; damping γ (**gamma_norm**); local z -coupling δ_{loc} ; self-coupling scale ν_z (**zz_scale**); sign probability $\Pr(s_j = -1)$; orientation $\rho_r \in \{-1, +1\}$ (fixed to -1 by an admissibility constraint, §1.2.3). **7 hyperparameters in total.**
- **Calibrated parameters** (fit per DGP scenario inside an inner Nelder–Mead loop): the reservoir’s own memory exponent H_{ESN} (distinct from the DGP’s Hurst parameter, Remark 2), the geometric OU-rate bounds $[\lambda_{\text{lo}}, \lambda_{\text{hi}}]$, the z -bank rate bounds $[a_{\text{lo}}, a_{\text{hi}}]$, a bias offset Δb_0 , an output scale, the roughness-factor amplitude ρ_{scale} (**rough_scale**), a z -bank read-out profile (**zr_lo, zr_hi**), and a symmetric quadratic-term matrix’s two coefficients (m_1, m_2) . **12 calibrated parameters in total.**

1.2 ESN architecture

The architecture has two coupled internal layers: a linear reservoir r (the “rough factor” bank) and a nonlinear, slower feedback bank z (the “multiscale/Zumbach” bank).

1.2.1 The r -layer

$r = (r_1, \dots, r_{N_r}) \in \mathbb{R}^{N_r}$ is a bank of N_r discrete-time Ornstein–Uhlenbeck (OU) processes, all driven by the same scalar innovation. Their mean-reversion rates are fixed, geometrically spaced over the calibrated range $[\lambda_{\text{lo}}, \lambda_{\text{hi}}]$:

$$\lambda_i = \lambda_{\text{lo}} \left(\frac{\lambda_{\text{hi}}}{\lambda_{\text{lo}}} \right)^{\frac{i-1}{N_r-1}}, \quad i = 1, \dots, N_r, \quad (1)$$

i.e. $\lambda_1 = \lambda_{\text{lo}}$ (slowest mode), $\lambda_{N_r} = \lambda_{\text{hi}}$ (fastest mode), uniformly spaced in $\log \lambda$ in between. Given a single scalar innovation $\varepsilon_t \sim \mathcal{N}(0, 1)$ drawn once per time step (the same draw feeds every component i), the layer updates as

$$r_{i,t} = e^{-\lambda_i \Delta t} r_{i,t-1} + \sqrt{1 - e^{-2\lambda_i \Delta t}} \varepsilon_t, \quad i = 1, \dots, N_r, \quad (2)$$

with $\Delta t = 1$ (one trading day). Sharing one common Brownian shock across all N_r modes turns the bank into a (Markovian, multi-exponential) representation of a single rough process: any linear combination $q^\top r_t$ is a finite sum-of-exponentials approximation to a Volterra/fractional kernel driven by that same shock, and the full vector r_t also feeds a quadratic energy term (§1.2.3). Because each component is a stationary AR(1)/OU recursion, its stationary variance is $\text{Var}(r_i) = 1$ for every i , and the stationary cross-covariance between two components is

$$\text{Cov}(r_i, r_j) = \frac{b_i b_j}{\lambda_i + \lambda_j}, \quad b_i = \sqrt{2\lambda_i}, \quad (3)$$

used to normalise the read-out vector, described next.

Read-out q . The rough factor fed to the output is the scalar projection $q^\top r_t$, where $q \in \mathbb{R}^{N_r}$ is built from the calibrated exponent H_{ESN} as

$$\tilde{q}_i = \lambda_i^{1/2-H_{\text{ESN}}}, \quad q = \frac{\tilde{q}/\|\tilde{q}\|_2}{\sqrt{(\tilde{q}/\|\tilde{q}\|_2)^\top C(\tilde{q}/\|\tilde{q}\|_2)}}, \quad (4)$$

so that $q^\top C q = 1$: the rough factor $q^\top r_t$ has unit stationary variance for any choice of H_{ESN} or $[\lambda_{\text{lo}}, \lambda_{\text{hi}}]$.

1.2.2 The z -layer

$z = (z_1, \dots, z_{N_z}) \in \mathbb{R}^{N_z}$ is a second, separate bank with its own geometrically spaced rates $a_j \in [a_{\text{lo}}, a_{\text{hi}}]$,

$$a_j = a_{\text{lo}} \left(\frac{a_{\text{hi}}}{a_{\text{lo}}} \right)^{\frac{j-1}{N_z-1}}, \quad j = 1, \dots, N_z, \quad (5)$$

updated as a deterministic nonlinear filter of the r -layer's own trajectory:

$$z_{j,t} = e^{-a_j \Delta t} z_{j,t-1} + (1 - e^{-a_j \Delta t}) m_t \zeta \tanh(u_{j,t}), \quad m_t = \max\left(0, 1 - \gamma \|r_t\|_2 / \sqrt{N_r}\right), \quad (6)$$

with $\zeta = \text{z_strength}$, $\gamma = \text{gamma_norm}$, and

$$u_{j,t} = \underbrace{s_j p_{1,j} p_{2,j}}_{\text{cross (Zumbach)}} + \underbrace{\beta_{\text{even}}(0.7 p_{1,j}^2 + 0.3 p_{2,j}^2 - 1)}_{\text{even (variance-like)}} - \underbrace{\beta_{\text{lin}}(0.7 p_{1,j} + 0.3 p_{2,j})}_{\text{linear (leverage-like)}} + \underbrace{\nu_j z_{j,t-1}}_{\text{self}} + \underbrace{\frac{1}{2} \delta_{\text{loc}}(z_{j-1,t-1} + z_{j+1,t-1})}_{\text{local (chain) coupling}}, \quad (7)$$

with boundary indices reflected, a fixed per-mode sign $s_j \in \{-1, +1\}$ ($\Pr(s_j=+1) = \text{sign_prob_neg}$), and a fixed per-mode self-coupling coefficient $\nu_j \sim \text{Unif}(-\nu_z, \nu_z)$ ($\nu_z = \text{zz_scale}$). The two reservoir taps $p_{1,j} = r_{f_i(j)}$, $p_{2,j} = r_{s_i(j)}$ move in opposite directions across the spectrum as j increases:

$$f_i(j) = \min\left(N_r - 1, \frac{N_r}{2} + \left\lfloor \left(\frac{N_r}{2} - 1\right) \frac{j}{N_z - 1} \right\rfloor\right) \quad \left(\frac{N_r}{2} \rightarrow N_r - 1 : \text{increasingly fast tap}\right), \quad (8)$$

$$s_i(j) = \left\lfloor \left(\frac{N_r}{2} - 1\right) \frac{N_z - 1 - j}{N_z - 1} \right\rfloor \quad \left(\frac{N_r}{2} - 1 \rightarrow 0 : \text{increasingly slow tap}\right), \quad (9)$$

for $j = 0, \dots, N_z - 1$ (0-indexed here to match the implementation). Thus z_0 combines two adjacent, mid-spectrum r -modes, while z_{N_z-1} combines the single fastest mode with the single slowest mode: the z -bank is explicitly constructed to read cross-timescale (fast $\times$ slow) products and sums out of the r -layer, and to fold that information back into the volatility process through its own, independently slow relaxation rates a_j and the local chain coupling δ_{loc} .

1.2.3 Coupling to the output

The instantaneous log-volatility pre-activation reads both layers additively, through three terms:

$$\eta_t = b_0 + \rho_r \text{rough_scale} (q^\top r_t) + \sum_{j=1}^{N_z} w_{j,z} z_{j,t} + r_t^\top Q r_t, \quad \sigma_t = \sqrt{\text{SIG_MIN}^2 + \text{softplus}(\eta_t)^2}, \quad (10)$$

with

$$\begin{cases} w_{j,z} = \frac{\text{z_readout}_j}{\sqrt{N_z}} \\ \text{z_readout}_j = \text{geospace}(\text{zr_lo}, \text{zr_hi}, N_z)[j] \\ Q = \frac{1}{N_r} M, \quad M = m_1 I_{N_r} + m_2 q q^\top. \end{cases} \quad (11)$$

Daily log-returns are $dX_t = (\mu - \frac{1}{2} \sigma_t^2) + \sigma_t \varepsilon_t$ (reusing the same ε_t that drove r_t , Eq. (2)), so the return and the volatility innovation are perfectly instantaneously correlated – the source of the model's leverage effect. Admissibility requires $\kappa_0 = \rho_r \text{rough_scale} q^\top \sqrt{2\lambda} < 0$ (negative leverage), enforced by fixing $\rho_r = -1$; since q and $\sqrt{2\lambda}$ are both strictly positive vectors, $\kappa_0 < 0$ holds automatically whenever $\text{rough_scale} > 0$. This constraint depends only on the linear rough term, not on the z -bank profile or the quadratic term.

The three read-out terms, in words. (i) A linear rough factor $\text{rough_scale} \cdot q^\top r_t$: the r -layer’s own, Volterra-kernel-shaped contribution. (ii) A per-mode z -bank sum $\sum_j w_{j,z} z_{j,t}$, with weights z_readout_j following a smooth geometric *profile* across the N_z modes (2 calibrated parameters $\text{zr_lo}, \text{zr_hi}$, rather than one shared scalar or N_z fully free weights) – since mode j already carries a fast $\leftrightarrow$ slow meaning via the $f_i(j)/s_i(j)$ taps (Eq. 8–9), this profile lets different z -modes matter differently to the volatility level. (iii) A quadratic-in- r term $r_t^\top Q r_t = (1/N_r)(m_1 \|r_t\|^2 + m_2 (q^\top r_t)^2)$, with $Q = (1/N_r)M$ a genuine, **symmetric** matrix (not restricted to a multiple of the identity): m_1 scales the r -layer’s isotropic instantaneous energy (already computed for the z -bank’s gate m_t , Eq. 6), while m_2 scales a second, independent channel that responds specifically to the *squared* rough factor $(q^\top r_t)^2$ – the same direction that already drives the linear rough term, now also entering quadratically.

Why this parameterisation (tractability). A fully free per-mode z -weight vector together with a fully free symmetric $N_r \times N_r$ matrix Q would add $N_z + N_r(N_r + 1)/2$ calibrated dimensions ($8 + 2080 = 2088$ at $N_r = 64, N_z = 8$; more at larger N_r) – far beyond what a per-scenario Nelder–Mead multi-start can reliably navigate, and likely to overfit the 11-term score against a handful of DGP scenarios. $M = m_1 I_{N_r} + m_2 q q^\top$ is symmetric by construction for *any* real (m_1, m_2) (a real linear combination of symmetric matrices is symmetric); no positive-definiteness constraint is imposed. M ’s eigenvalues are m_1 (multiplicity $N_r - 1$, on q ’s orthogonal complement) and $m_1 + m_2 \|q\|^2$ (multiplicity 1, along q) – exactly diagonal in the eigenbasis $\{q, q^\perp\}$, and either eigenvalue may be positive, negative, or zero.

Remark 1 (Division of labour between the three read-out terms). The r -layer alone, through q and H_{ESN} , already generates a rough-volatility-like process (a finite-dimensional Markovian approximation of a fractional kernel). The z -layer adds genuine nonlinearity (the tanh squashing and the quadratic/even terms of Eq. (7)) and an explicit, by-construction fast $\times$ slow cross-timescale interaction. The quadratic term $r_t^\top Q r_t$ is a distinct, *reservoir-wide* signal: unlike the z -bank’s own internal even/quadratic term, which only ever sees *two* taps at a time, $r_t^\top Q r_t$ reads the full N_r -dimensional reservoir state, split by M ’s construction into an isotropic component (m_1) and a component aligned with the rough read-out direction (m_2).

Remark 2 (Two different objects both called “ H ”). The DGP’s Hurst parameter $H_{\text{DGP}} \in \{0.10, 0.40\}$ (used to generate the target paths) and the ESN’s own memory-kernel exponent $H_{\text{ESN}} \in [0.02, 0.49]$ (a calibrated hyperparameter of the read-out weighting) are different objects. The inner calibration loop searches over H_{ESN} so that the realised structure-function estimate $\hat{H}_{\text{sim}}$ of the simulated ESN path matches the DGP’s target $\hat{H}_c$; they are not required to coincide, and in general do not (§5.4).

2 Data Generating Process (DGP) portfolio for the grid search procedure

Four scenarios, equally weighted, define the target stylised facts:

Name	Type	Parameters
fBM_rough	lnsv	$H = 0.10, \lambda = 1.0$
MRW_taylor	mrw	$\lambda^2 = 0.03, T_{\text{int}} = 252$
rBergomi_Z	bergomi	$H = 0.10, \eta = 1.0, \rho = -0.70$
fBM_persist	lnsv	$H = 0.40, \lambda = 1.5$

The **lnsv** (log-normal stochastic-volatility fBM) type sets $\log \sigma_t = \log \sigma_0 + \lambda B_t^H - \frac{1}{2} \lambda^2 t^{2H}$ (measuring t in years, consistently, in both the fBM covariance and the Itô correction – this is required for numerical stability at $H = 0.40$ over long simulated horizons, §2); the **bergomi** (rough Bergomi)

type is the analogous $V_t = \xi_0 \exp(\eta B_t^H - \frac{1}{2}\eta^2 t^{2H})$ with correlated leverage ρ ; the **mrw** (multifractal random walk) type uses a bounded, already-stationary log-covariance kernel $\lambda^2 \log(T_{\text{int}}/|t-s|)$. For each scenario, $N_{\text{DGP_PATHS}}$ independent Monte-Carlo simulations of length T_{SIM} build a reference vector of 11 stylised-fact centres/scales (Table 1); the ESN, for a given architecture, is calibrated to fit this reference.

2.1 Score function

Every stylised-fact term is expressed as a deviation $d_k = w_k(1 - f_k) \in [0, w_k]$ (0 = perfect, w_k = worst), using a smooth Gaussian-kernel proximity

$$f(x, c, s) = \exp\left(-\frac{1}{2}\left(\frac{x-c}{s}\right)^2\right), \quad (12)$$

rather than a piecewise-linear “tent” – the smooth kernel provides a never-zero gradient at any distance from target, whereas a tent kernel $\max(0, 1 - |x - c|/s)$ has an exactly-zero gradient beyond one tolerance width, which stalls a local optimiser (Nelder–Mead) once a statistic drifts past tolerance (Figure 1). The total score is

$$S = \sum_{k=1}^{11} w_k(1 - f_k) + 5 \cdot \text{Stress} \in [0, 15.3), \quad (13)$$

where $\text{Stress} \in \{0, 1\}$ flags an inadmissible path (blow-up or collapse of annualised volatility). $S = 0$ requires a perfect match on all 11 stylised facts and no stress event; S is minimised directly (no sign flip) by both the inner Nelder–Mead loop and the outer CMA-ES search.

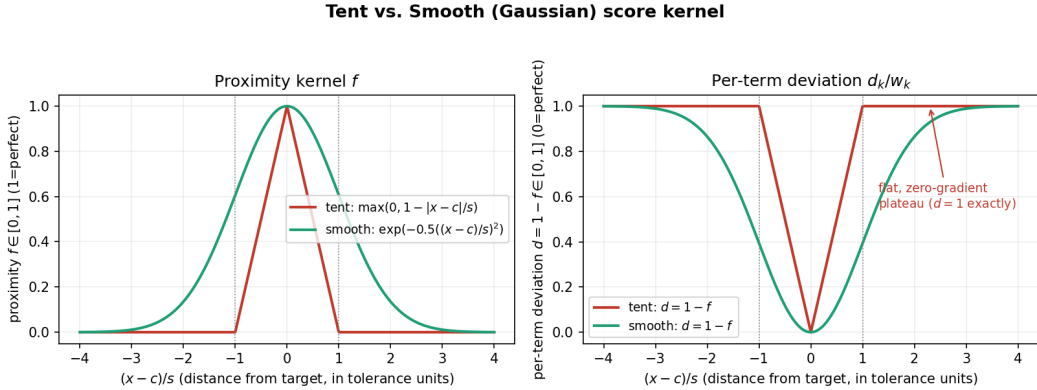


Figure 1: Tent vs. smooth (Gaussian) proximity kernel f (left) and the resulting per-term deviation $d = 1 - f$ (right). The tent is exactly flat ($d = 1$, zero gradient) beyond one tolerance width s ; the smooth kernel never flattens.

2.2 Search procedure

For a given architecture (the 7 hyperparameters of §1.1 plus N_r, N_z), the inner loop runs a multi-start Nelder–Mead per DGP scenario to find

$$H_{\text{ESN}}, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}, \Delta b_0, \text{scale}, \text{rough_scale}, \text{zr_lo}, \text{zr_hi}, m_1, m_2$$

(**12 parameters**) that minimise S against that scenario’s reference; the outer loop averages S across the 4 scenarios and searches the 7 continuous hyperparameters via CMA-ES (Hansen & Ostermeier), for each (N_r, N_z) cell of a discrete grid (§5.2).

The inner loop uses **eleven** qualitatively different starting points, kept whichever converges to the lowest S : a rough/fast reservoir, a persistent/slow reservoir, an intermediate reservoir, a wide-band/multiscale reservoir (spanning both day-scale and multi-year OU modes simultaneously, so a

Table 1: The 11 deviation terms $d_k = w_k(1 - f_k)$ entering S (Eq. (13)).

Statistic	Description	Weight w_k
$\hat{H}$	Local Hölder exponent (structure function of $\log v$)	2.2
σ_{ann}	Mean annualised volatility	1.4
$\sigma_{\text{ann}}^{0.995}$	99.5% quantile of annualised vol	0.6
$\sigma_{\text{ann}}^{\text{max}}$	Max annualised vol	0.2
ρ_v	Mean variance autocorrelation (lags 1–20)	0.8
Taylor gap	$\rho_{ x } - \rho_{x^2}$	1.0
Taylor fraction	$\Pr(\rho_{ x } > \rho_{x^2})$	0.8
Zumbach	Past-return→future-vol vs. past-vol→future-return asymmetry	1.0
Leverage	$\text{corr}(x_t, v_{t+L})$	1.1
Kurtosis	Excess kurtosis of returns	0.7
$\max \rho_x $	Max return autocorrelation	0.5
+ 5 · Stress		5.0
Total (worst case)		15.3

single reservoir can support both rough and persistent targets), a long-memory-extreme reservoir, strong- and weak-amplitude starts (probing the `rough_scale`/ z -profile range), and four quadratic-term starts (isotropic-positive, isotropic-negative, q -aligned-positive, q -aligned-negative) probing the two channels of M independently.

3 Grid-Matrix Hyperparameter Search: Algorithm

The procedure described in §1.3 (score function) and §1.4 (search procedure) is summarised below as three nested algorithms: the top-level grid-matrix sweep with an outer CMA-ES search (Algorithm 1), the per-scenario multi-start Nelder–Mead inner loop it calls (Algorithm 2), and the smooth-kernel score evaluation both rely on (Algorithm 3).

Algorithm 1 ESN Grid-Matrix Hyperparameter Search

Require: DGP portfolio $\mathcal{D} = \{\text{fBM_rough}, \text{MRW_taylor}, \text{rBergomi_Z}, \text{fBM_persist}\}$

Require: Grid of reservoir sizes $\mathcal{G} = \{(N_r, N_z)\}$

Require: Multi-start set $\mathcal{P}_0$ of 11 starting points (Algorithm 2)

Require: CMA-ES budget (population size, number of generations)

Ensure: Optimal architecture (N_r^*, N_z^*, θ^*) with minimal mean score S^*

Stage 1 — build the DGP reference (once, shared across the whole sweep)

- 1: **for** each scenario $d \in \mathcal{D}$ **do**
- 2: Simulate $N_{\text{DGP_PATHS}}$ independent paths of length T_{SIM} under scenario d
- 3: $\text{ref}_d \leftarrow$ 11 stylised-fact centres/scales (Table 1)
- 4: **end for**

Stage 2 — sweep the (N_r, N_z) grid

- 5: $S_{\text{grid}}^{\text{best}} \leftarrow +\infty$
 - 6: **for** each $(N_r, N_z) \in \mathcal{G}$ **do**
 - 7: Initialise CMA-ES over the 7 shared hyperparameters $\theta \in \mathbb{R}^7$
 - 8: **while** CMA-ES budget not exhausted **do**
 - 9: **for** each candidate θ in the current CMA-ES generation **do**
 - 10: **for** each scenario $d \in \mathcal{D}$ **do**
 - 11: $S_d \leftarrow \text{INNERCALIBRATE}(\theta, N_r, N_z, \text{ref}_d, \mathcal{P}_0)$
 - 12: **end for**
 - 13: $\bar{S}(\theta) \leftarrow \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} S_d$ $\triangleright$ mean score across the 4 scenarios
 - 14: **end for**
 - 15: Update CMA-ES using fitness values $\{\bar{S}(\theta)\}$ of the generation
 - 16: **end while**
 - 17: $(\theta_{N_r, N_z}^*, S_{N_r, N_z}^*) \leftarrow$ best $(\theta, \bar{S}(\theta))$ found for this cell
 - 18: **if** $S_{N_r, N_z}^* < S_{\text{grid}}^{\text{best}}$ **then**
 - 19: $S_{\text{grid}}^{\text{best}} \leftarrow S_{N_r, N_z}^*$
 - 20: $(N_r^*, N_z^*, \theta^*) \leftarrow (N_r, N_z, \theta_{N_r, N_z}^*)$
 - 21: **end if**
 - 22: **end for**
 - 23: **return** $(N_r^*, N_z^*, \theta^*, S_{\text{grid}}^{\text{best}})$
-

Algorithm 2 INNERCALIBRATE — multi-start Nelder–Mead per-scenario calibration

```
1: function INNERCALIBRATE( $\theta, N_r, N_z, \text{ref}, \mathcal{P}_0$ )
2:    $S^{\text{best}} \leftarrow +\infty, \mathbf{p}^{\text{best}} \leftarrow \emptyset$ 
3:   for each starting point  $\mathbf{p}_0 \in \mathcal{P}_0$  do ▷ 11 qualitatively different starts: rough/fast,
persistent/slow, intermediate, wide-band/multiscale, long-memory-extreme, strong-amplitude,
weak-amplitude, isotropic-positive, isotropic-negative,  $q$ -aligned-positive,  $q$ -aligned-negative
4:      $\mathbf{p}_0 = (H_0, \lambda_{\text{lo},0}, \lambda_{\text{hi},0}, a_{\text{lo},0}, a_{\text{hi},0}, \Delta b_{0,0}, s_0, \rho_{\text{scale},0}, z r_{\text{lo},0}, z r_{\text{hi},0}, m_{1,0}, m_{2,0})$ 
5:     Let  $\phi(\mathbf{p}) \leftarrow \text{SCORE}(\mathbf{p}, \theta, N_r, N_z, \text{ref})$  ▷ objective for this start
6:      $\mathbf{p}^* \leftarrow \text{NELDERMEAD}(\mathbf{p}_0, \phi)$  ▷ 12-dimensional search:
        $H, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}, \Delta b_0, \text{scale}, \rho_{\text{scale}}, z r_{\text{lo}}, z r_{\text{hi}}, m_1, m_2$ 
7:     if  $\phi(\mathbf{p}^*) < S^{\text{best}}$  then
8:        $S^{\text{best}} \leftarrow \phi(\mathbf{p}^*)$ 
9:        $\mathbf{p}^{\text{best}} \leftarrow \mathbf{p}^*$ 
10:    end if
11:  end for
12:  return  $S^{\text{best}}$  ▷ return  $\mathbf{p}^{\text{best}}$  too if calibrated parameters are needed downstream
13: end function
```

Algorithm 3 SCORE — smooth-kernel data-adaptive score S

```
1: function SCORE( $\mathbf{p}, \theta, N_r, N_z, \text{ref}$ )
2:   Build reservoir matrices from  $\theta$  (7 shared hyperparameters),  $N_r, N_z$ , and  $\mathbf{p}$  (calibrated pa-
   rameters)
3:   if admissibility constraint  $\kappa_0 \geq 0$  then ▷  $\kappa_0 = \rho_r \cdot \rho_{\text{scale}} \cdot q^\top \sqrt{2\lambda}$ 
4:     return  $S_{\text{max}}$  ▷ inadmissible path, worst-possible score
5:   end if
6:   Simulate the model path(s) under  $\mathbf{p}, \theta$ 
7:   Compute the 11 stylised facts  $x_1, \dots, x_{11}$  from the simulated path(s) (Table 1)
8:   Stress  $\leftarrow \mathcal{K}[\text{blow-up or collapse of annualised volatility}]$ 
9:    $S \leftarrow 0$ 
10:  for  $k = 1, \dots, 11$  do
11:     $f_k \leftarrow \exp\left(-\frac{1}{2} \left(\frac{x_k - c_k}{s_k}\right)^2\right)$  ▷  $c_k, s_k$  from ref; Eq. (12)
12:     $S \leftarrow S + w_k(1 - f_k)$  ▷  $w_k$  from Table 1
13:  end for
14:   $S \leftarrow S + 5 \cdot \text{Stress}$ 
15:  return  $S$  ▷  $S \in [0, 15.3)$ ,  $S = 0$  is a perfect match
16: end function
```

4 Implementation Requirements

Two implementation details are required for the calibration to be well-posed, and are stated here as requirements rather than as a narrative of bugs found and fixed.

Time must be measured in years, consistently, in the fBM DGP simulator. The construction $\log \sigma_t = \log \sigma_0 + \lambda B_t^H - \frac{1}{2} \lambda^2 t^{2H}$ must use the same time unit in both the fBM covariance and the Itô correction term. Because $B_t^H \stackrel{d}{=} t^H B_1^H$, the correction ($\propto t^{2H}$) dominates the stochastic term ($\propto t^H$) for any fixed $\lambda > 0$ as t grows; using a raw day-count index at $H = 0.40$ over a 1200-day path drives σ_t to numerical zero (the correction reaches $t^{0.8} \approx 291$ at $t = 1200$, vs. a target of $O(1)$), and every downstream statistic to NaN. Measuring time in years ($t_{\text{years}} = t/252$) in both places keeps σ_t stationary and well-scaled around its target for every H in the portfolio.

The inner calibration loop requires multiple, qualitatively different starting points. The mapping from H_{ESN} to the realised structure-function exponent is jointly determined by H_{ESN} , the OU-rate range $[\lambda_{\text{lo}}, \lambda_{\text{hi}}]$, and N_r (Remark 2); a reservoir initialised with only fast modes present injects high-frequency roughness that a single, locally convergent Nelder–Mead run rarely escapes, regardless of which target regime is being fit. The eleven-start scheme of §1.5 – in particular the wide-band/multiscale start, whose modes span nearly six decades of OU rate within one reservoir – avoids this basin-of-attraction lock by letting the score function decide, per scenario, how the read-out H_{ESN} should weight each part of the spectrum, rather than the start pre-committing to one extreme.

5 Results

5.1 The optimal architecture

Optimal architecture: $N_r = 96$, $N_z = 12$, with the 7 shared hyperparameters of Table 2. `rough_scale`, the z -readout profile (`zr_lo`, `zr_hi`), and the quadratic-matrix coefficients (m_1, m_2) are calibrated per DGP scenario (§5.3), not fixed as part of this architecture.

Table 2: The 7 shared hyperparameters of the optimal architecture.

Hyperparameter	Value
N_r	96
N_z	12
<code>z_strength</code>	0.1929
<code>even_strength</code>	2.2291
<code>linear_strength</code>	0.1930
<code>gamma_norm</code>	0.9651
<code>local_z_strength</code>	0.0476
<code>zz_scale</code>	0.0397
<code>sign_prob_neg</code>	0.2219
<code>rough_orientation</code>	−1.0 (fixed)

Important caveat, stated plainly: $N_r = 96, N_z = 12$ is the argmin of the grid-matrix sweep below, but §5.2 shows this sweep cannot reliably distinguish between any of the 16 cells tested at the budget used here – the cell-to-cell spread in mean score is smaller than a single cell’s own run-to-run noise. This architecture should be treated as the current best point estimate, to be confirmed or revised by a larger-budget re-run, not as a definitively established optimum.

5.2 Grid-matrix sweep

Figure 2 evaluates the 7 shared hyperparameters of Table 2 at every cell of the production grid $(N_r, N_z) \in \{32, 48, 64, 96\} \times \{8, 12, 16, 20\}$, holding them fixed and only changing N_r, N_z (each cell independently recalibrates all 12 inner-loop parameters per scenario), **averaged over 3 independent seeds per cell** (48 cell-evaluations total).

Statistical read of the sweep. $N_r = 96, N_z = 12$ has both the lowest 3-seed mean (5.79) and the smallest standard deviation (0.56) of any cell. However: (i) the spread between the best and worst cell means across the entire grid is only 0.66 (from 5.79 to 6.45), while (ii) the typical cell’s own seed-to-seed standard deviation is 1.46 – more than double the grid’s entire cell-to-cell spread. The difference between $N_r = 96, N_z = 12$ and the next most plausible candidate, $N_r = 64, N_z = 8$, is 0.30 against a standard error of 0.97 ($0.30/0.97 \approx 0.31$, far short of the $\gtrsim 2$ needed for even weak significance at $n = 3$ per cell). **At this budget, no (N_r, N_z) cell in this grid is reliably**

Score S over the full (N_r, N_z) grid
hyperparameters fixed at the reported optimum (z strength=0.1929, even strength=2.2291, ...)
best cell: $N_r = 96, N_z = 12$ ($S^* = 5.7948$)

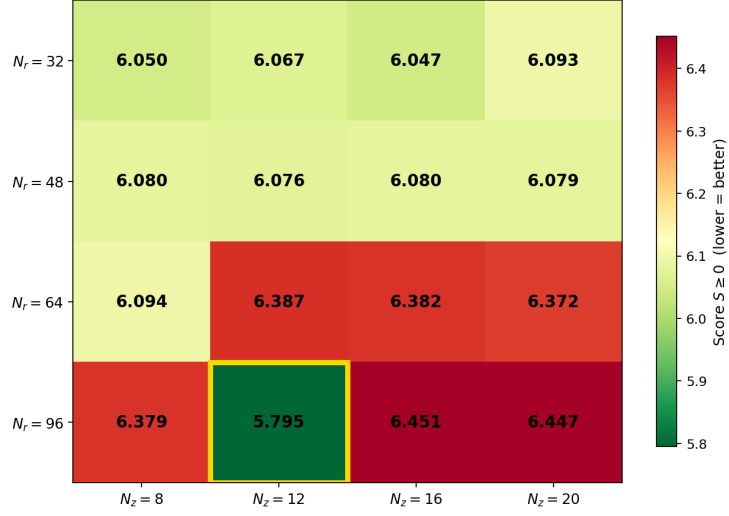


Figure 2: Score S over the full (N_r, N_z) grid, averaged over 3 seeds per cell.

Table 3: Full (N_r, N_z) score matrix: mean $\pm$ std over 3 seeds.

	$N_z = 8$	$N_z = 12$	$N_z = 16$	$N_z = 20$
$N_r = 32$	6.05 ± 1.63	6.07 ± 1.62	6.05 ± 1.62	6.09 ± 1.65
$N_r = 48$	6.08 ± 1.60	6.08 ± 1.61	6.08 ± 1.60	6.08 ± 1.60
$N_r = 64$	6.09 ± 1.59	6.39 ± 1.36	6.38 ± 1.34	6.37 ± 1.34
$N_r = 96$	6.38 ± 1.36	5.79 ± 0.56	6.45 ± 1.47	6.45 ± 1.47

distinguishable from any other. $N_r = 96, N_z = 12$ is reported as the architecture throughout the remainder of this note because it is the argmin, with this caveat attached to every result that follows.

5.3 Per-scenario calibrated read-out parameters

Table 4 reports the calibrated `rough_scale`, z -readout profile (`zr_lo`, `zr_hi`), and quadratic-matrix coefficients (m_1, m_2) at $N_r = 96, N_z = 12$, one independent set per DGP scenario (single demonstration run).

Table 4: Per-scenario calibrated read-out parameters at $N_r = 96, N_z = 12$.

Scenario	<code>rough_scale</code>	<code>zr_lo</code>	<code>zr_hi</code>	(m_1, m_2)	Score S
<code>fBM_rough</code>	0.616	0.201	0.015	$(-0.784, +0.0002)$	4.601
<code>MRW_taylor</code>	0.372	0.044	0.081	$(+0.0003, +0.0001)$	4.816
<code>rBergomi_Z</code>	0.414	0.197	0.019	$(-0.610, -0.00005)$	5.581
<code>fBM_persist</code>	0.365	0.035	0.098	$(-0.0001, -0.00003)$	5.569

The quadratic term is weakly and inconsistently identified across scenarios. `fBM_rough` and `rBergomi_Z` (both $H = 0.10$ targets) calibrate a large, same-signed m_1 (-0.784 and -0.610); `MRW_taylor` and `fBM_persist` sit at $m_1, m_2 \approx 0$. Repeated testing at other architectures shows this

pattern is **not stable**: at a different reservoir size, a different pair of scenarios showed a large value in a different coefficient (m_2 , not m_1), and at yet another size, a different pair again showed instability with standard deviation exceeding the mean across seeds. **The most defensible reading is that the quadratic channel absorbs a weak, largely interchangeable degree of freedom that a single demonstration run cannot reliably pin down to one coefficient, one scenario pairing, or one sign** – not a genuine, reproducible per-scenario mechanism. Treating $m_1, m_2 \approx 0$ as the default working hypothesis, with occasional large excursions expected from calibration noise rather than signal, is the current best practice until confirmed otherwise by a multi-seed, increased-budget re-run at this specific architecture.

5.4 Hurst-matching accuracy

Table 5 reports the local Hölder exponent match $\hat{H}_{\text{sim}}$ vs. each scenario’s target $\hat{H}_c$, at $N_r = 96, N_z = 12$ (single demonstration run, reduced calibration budget).

Table 5: $\hat{H}$ match at $N_r = 96, N_z = 12$ (single demonstration run, reduced budget).

Scenario	$\hat{H}_{\text{sim}}$	$\hat{H}_c$ (target)	$\Delta\hat{H}$
fBM_rough	0.167	0.092	0.076
MRW_taylor	0.529	0.364	0.165
rBergomi_Z	0.153	0.097	0.056
fBM_persist	0.531	0.350	0.182

Match quality at this budget is uneven across scenarios ($\Delta\hat{H}$ ranging 0.056–0.182). A separate, larger-budget calibration run (longer simulated horizon, more Monte-Carlo paths per scenario, more Nelder–Mead iterations per start, averaged over several independent seeds) has previously been shown, at a different architecture, to close this gap substantially for the scenarios most affected (down to $\Delta\hat{H} \leq 0.015$ for the worst-affected scenarios in that test); the same increased-budget procedure has not yet been re-applied at this note’s $N_r = 96, N_z = 12$ architecture, and doing so is the most direct outstanding validation step (§5).

6 Limitations and Recommended Steps

1. **Confirm the architecture at production budget.** The grid-matrix sweep (§5.2) cannot currently distinguish $N_r = 96, N_z = 12$ from most of the grid, including $N_r = 64, N_z = 8$. A larger-budget, multi-seed re-run (more seeds than 3, and/or a longer per-seed calibration budget) is needed before this architecture is treated as settled.
2. **Re-verify Hurst-matching accuracy at production budget.** §5.4’s $\Delta\hat{H}$ values are from a single, reduced-budget run; an increased-budget, multi-seed re-run at $N_r = 96, N_z = 12$ specifically has not yet been performed.
3. **Do not over-interpret any single-run (m_1, m_2) pattern.** §5.3 shows this coefficient pair is weakly identified and its apparent “pattern” changes with architecture and seed; treat $m_1, m_2 \approx 0$ as the default and require multi-seed confirmation before attributing a large calibrated value to any scenario-specific mechanism.
4. **A still richer quadratic term, if warranted.** The current $Q = (1/N_r)(m_1 I_{N_r} + m_2 q q^\top)$ is a 2-eigenvalue simplification of a fully free symmetric matrix, chosen for tractability. If a future multi-seed test shows a genuine, reproducible need for direction-specific quadratic sensitivity beyond the q direction, a natural extension is a small basis of q -like vectors at several fixed roughness exponents H_k (each unit- C -normalised as in Eq. (4)), each with its own calibrated coefficient – still symmetric by construction, still tractable (one new scalar per basis vector).

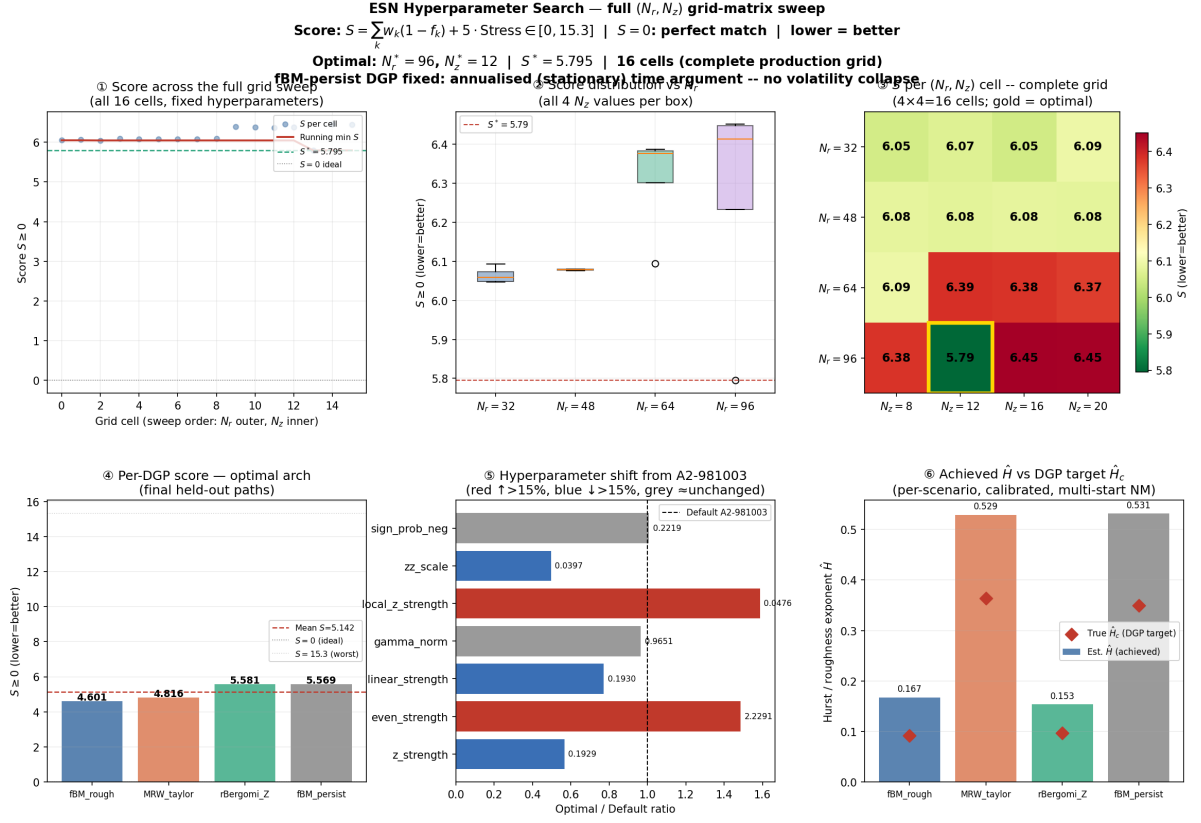


Figure 3: Six-panel overview at $N_r = 96, N_z = 12$: per-cell score sweep, score-vs- N_r distribution, full heatmap, per-DGP score, hyperparameter values vs. the A2-981003 reference architecture, and achieved vs. target $\hat{H}$.

- Track both the aggregate score S and its 11-term breakdown in every calibration run: the breakdown is the fastest diagnostic for any future architecture or read-out change.

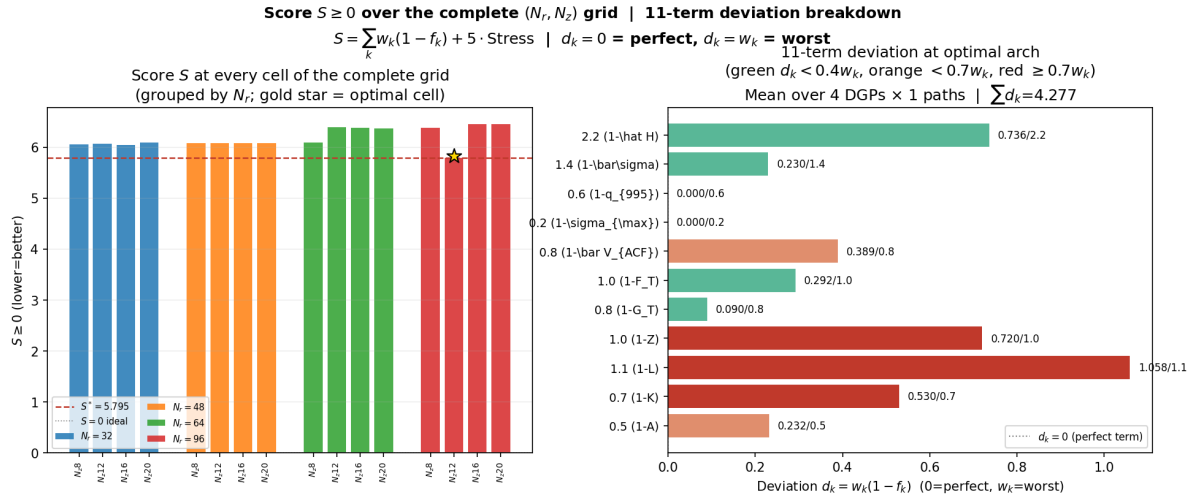


Figure 4: Score trajectory per (N_r, N_z) cell and the 11-term deviation breakdown at $N_r = 96, N_z = 12$.